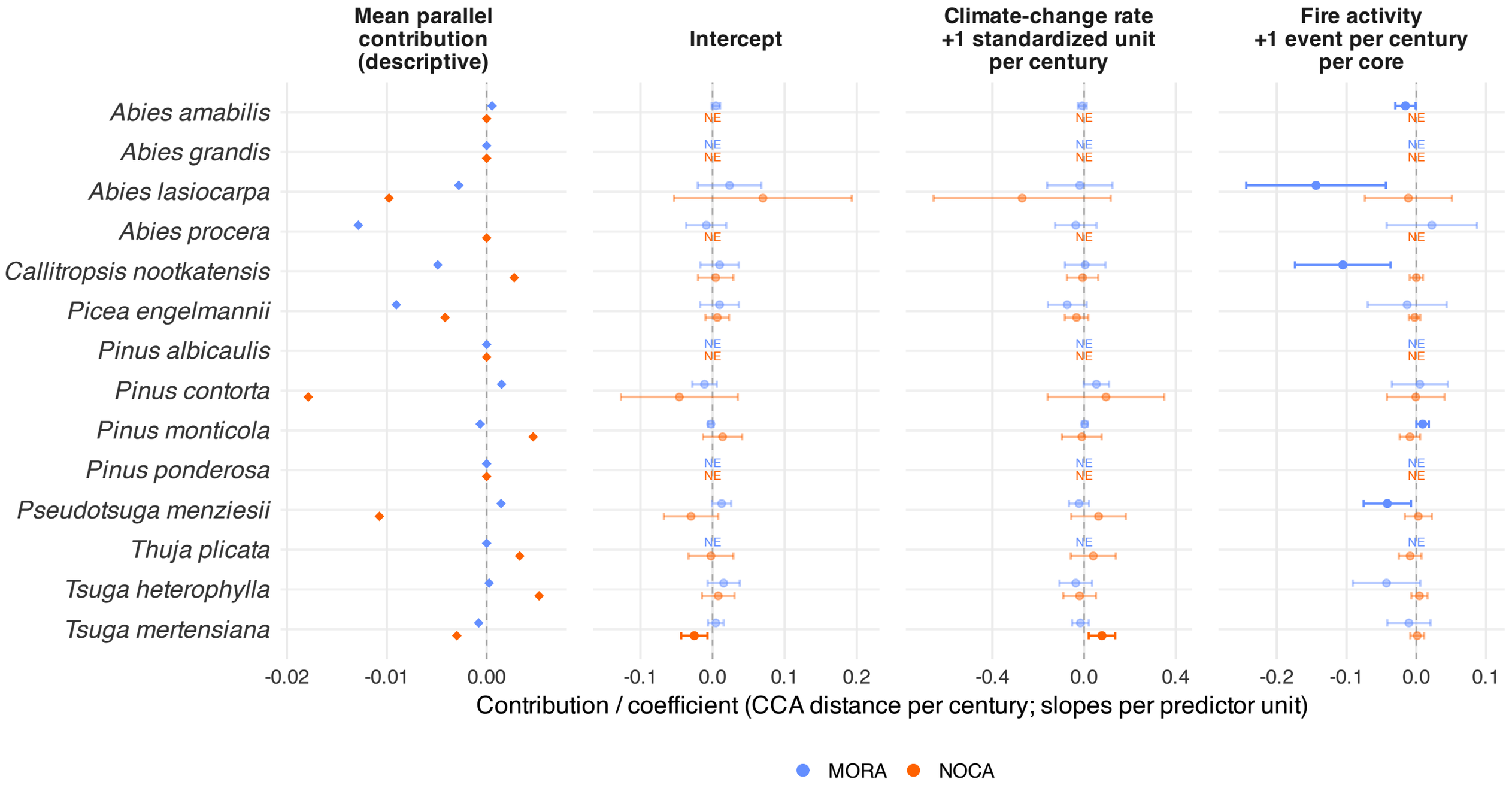
Holocene

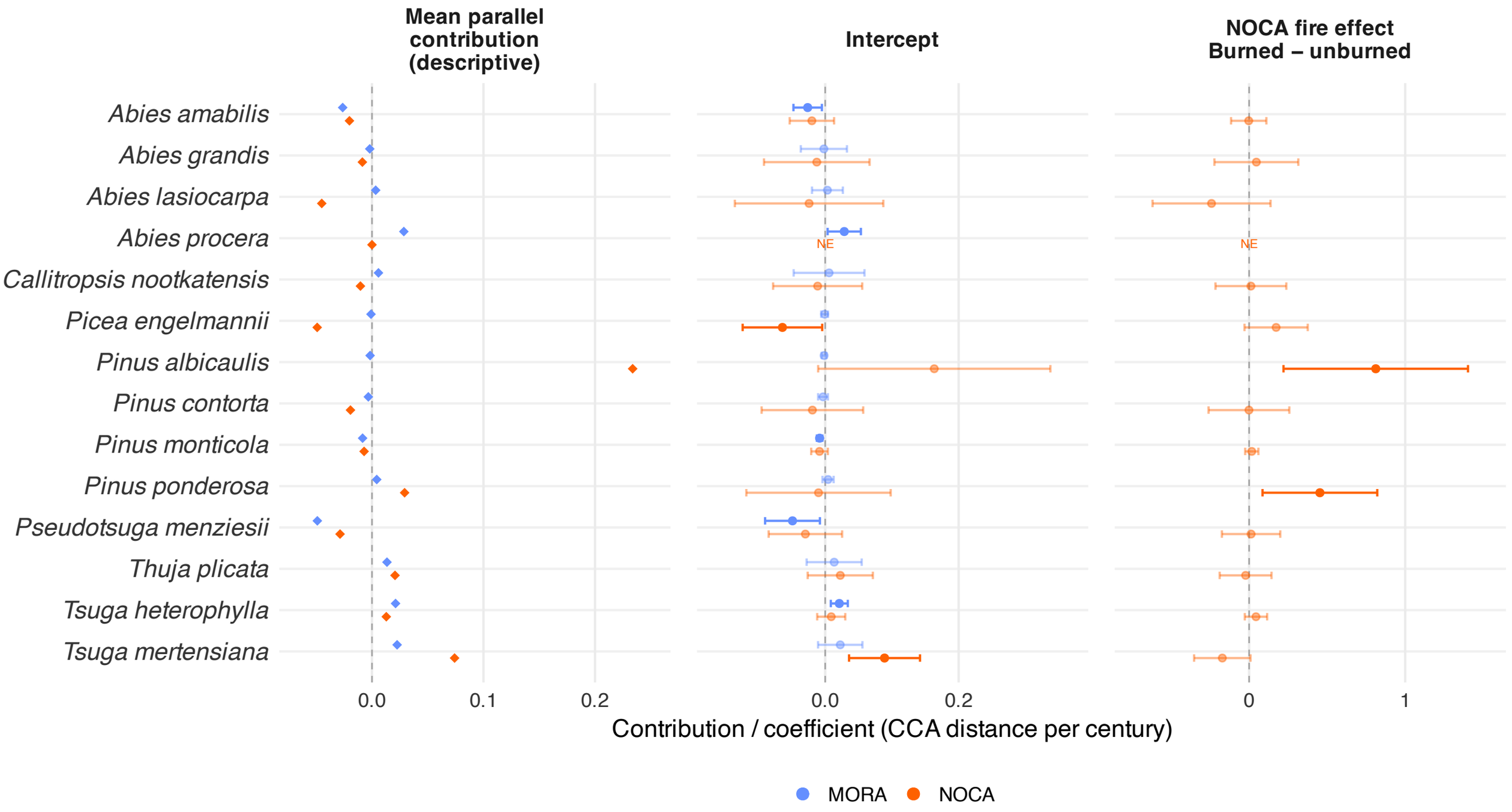
Within each park: contribution rate ~ climate-change rate + fire activity · MA(1)



Diamonds: arithmetic means of all finite primary contributions (the boxplot data); each transition/plot enters once, with no weighting.
Means retain zeros and omit undefined directions; they have no model confidence intervals or significance fading.
Circles: separate park GLS estimates; bars: conditional 95% confidence intervals. Faded estimates: unadjusted $p \geq 0.05$.
NE = not estimable in that park; absence in the other park does not prevent a fit. Each coefficient column has its own x-axis scale.
Intercept: expected contribution in that park at zero climate-change rate and zero fire activity; it is not the overall mean.
Positive contributions align with climate-predicted movement; negative contributions oppose it. An intercept does not fix the sign of every observation.
Absent in MORA: Abies grandis, Pinus albicaulis, Pinus ponderosa, Thuja plicata.
Absent in NOCA: Abies amabilis, Abies grandis, Abies procera, Pinus albicaulis, Pinus ponderosa.

Anthropocene

MORA: contribution rate ~ 1 · NOCA: contribution rate ~ burned · Independent residuals



Diamonds: arithmetic means of all finite primary contributions (the boxplot data); each transition/plot enters once, with no weighting.
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Circles: separate park GLS estimates; bars: conditional 95% confidence intervals. Faded estimates: unadjusted $p \geq 0.05$.
NE = not estimable in that park; absence in the other park does not prevent a fit. Each coefficient column has its own x-axis scale.
MORA intercept: mean across all eligible plots. NOCA intercept: unburned-plot mean; fire effect: burned minus unburned. No MORA fire term is fitted.
Positive contributions align with climate-predicted movement; negative contributions oppose it. An intercept does not fix the sign of every observation.
Absent in NOCA: Abies procera.